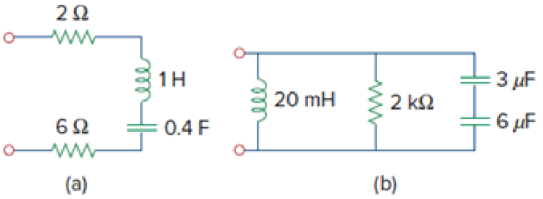
Chapter 14, Problem 42P

Problem

For the circuits in Fig. 14.81, find the resonant frequency $\omega0$, the quality factor Q, and the bandwidth B.

Figure 14.81:



Step-by-step solution

Step 1 of 2

(a) This is series RLC circuit

$$R = 2 + 6 = 8\Omega$$

$$L = 1H$$

$$C = 0.4F$$

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.4}} = 1.5811\, rad/s$$

$$\theta = \frac{\omega_o L}{R} = \frac{1.5811}{8} = 0.1976$$

$$B = \frac{R}{L} = 8 rad/s$$

Step 2 of 2

(b) This is a parallel RLC circuit

$$3\mu\text{F and } 6\mu\text{F} \rightarrow \frac{3\times 6}{3+6} = 2\mu F$$

$$C = 2\mu F \qquad R = 2K\Omega \qquad L = 20MH$$

$$\omega_o = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\left(2\times 10^{-6}\right)\left(20\times 10^{-3}\right)}} = 5K\, rad/s$$

$$\theta = \frac{R}{\omega_o L} = \frac{2\times 10^{3}}{\left(5\times 10^{3}\right)\left(20\times 10^{-3}\right)} = 20$$

$$B = \frac{1}{RC} = \frac{1}{\left(2\times 10^{3}\right)\left(2\times 10^{-6}\right)} = 250 rad/s$$